Lesson 7.7 — Graphing Trigonometric Functions

Big idea: To sketch $y = a \cdot f(b(x - h)) + k$ for sin, cos, or tan: identify the four parameters, then plot key points (max, min, zeros) on the first cycle.

Quick review

Step 1: identify a, b, h, k.

Step 2: compute amplitude ($|a|$), period ($2\pi/|b|$ for sin/cos, $\pi/|b|$ for tan), midline ($y = k$).

Step 3: mark phase shift on the x-axis (start of the cycle is at $x = h$).

Step 4: divide one period into four equal pieces; plot key points at each (max/zero/min/zero for sine, max/zero/min/zero for cosine, asymptotes/zero for tangent).

Step 5: connect smoothly and repeat the pattern.

Worked example

Worked example. Sketch $y = 2 \sin(2x - \pi) + 1$.

Rewrite as $y = 2 \sin(2(x - \pi/2)) + 1$. Amplitude 2, period π, phase shift right $\pi/2$, midline $y = 1$.
Key points (one cycle): start $(\pi/2, 1)$, max $(\pi/2 + \pi/4, 3) = (3\pi/4, 3)$, zero $(\pi, 1)$, min $(5\pi/4, -1)$, zero $(3\pi/2, 1)$.

Warm-ups

1. List five key x-values for one cycle of $y = \sin x$ starting at $x = 0$.
2. What is the period of $y = 3 \sin(4x)$?
3. What is the midline of $y = \cos x - 2$?

Practice

1. Sketch (describe) $y = 3 \cos(\pi x)$ for one cycle.
2. Sketch (describe) $y = \sin(x - \pi/2) - 1$ for one cycle.
3. Sketch (describe) $y = \tan(x)$. Identify asymptotes for $-\pi/2 < x < 3\pi/2$.

Challenge

1. Sketch $y = 2 \sin((\pi/3)x - \pi/3) + 1$. Find amplitude, period, phase shift, midline, and one full cycle of key points.
2. Sketch $y = -3 \cos(2x) + 4$. State all relevant parameters and the maximum/minimum values.

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Answer key — Lesson 7.7

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $0, \pi/2, \pi, 3\pi/2, 2\pi$.

2. $2\pi/4 = \pi/2$.

3. $y = -2$.

Practice

1. Amplitude 3, period 2. Cycle: $(0, 3) \to (1/2, 0) \to (1, -3) \to (3/2, 0) \to (2, 3)$.

2. Amplitude 1, period 2π, phase shift right $\pi/2$, midline $y = -1$. Cycle: $(\pi/2, -1) \to (\pi, 0) \to (3\pi/2, -1) \to (2\pi, -2) \to (5\pi/2, -1)$.

3. Period π. Asymptotes at $x = -\pi/2, \pi/2, 3\pi/2$. Crosses zero at $x = 0$ and $x = \pi$. Increasing on each interval between asymptotes.

Challenge

1. Rewrite: $y = 2 \sin((\pi/3)(x - 1)) + 1$. Amplitude 2. Period $2\pi/(\pi/3) = 6$. Phase shift right 1. Midline $y = 1$. One cycle from $x = 1$: max $(1 + 6/4, 3) = (5/2, 3)$, zero $(4, 1)$, min $(11/2, -1)$, zero $(7, 1)$.

2. Amplitude 3, period π, midline $y = 4$. Reflected ($a = -3 < 0$). Max value $= 4 + 3 = 7$. Min value $= 4 - 3 = 1$.

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